

CONSTRAINTS ON THE EMERGENCE AND
INTERPRETABILITY OF LINGUISTIC CONVENTIONS

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I certify that I have read this dissertation and that, in my opinion, it is fully adequate in scope and quality as a dissertation for the degree of Doctor of Philosophy.

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Abstract

Human communication takes place across many circumstances, and humans are adept at flexibly using language to achieve goals even in constrained conditions. The formation of linguistic conventions is often studied in iterated reference games, where over repeated reference to the same targets, a describer–matcher pair establishes partner-specific shorthand names for targets. In this dissertation, we expand the range of empirical evidence about when conventionalization occurs and what types of conventions are formed, all within the framework of iterated reference games. In Chapter 2, we ask which properties of the group’s interaction structure facilitate successful communication, using a repeated reference game paradigm where we manipulated several key constraints on the group’s interaction, including group size, the amount of feedback that matchers could give to directors, and the availability of peer interaction between matchers. In Chapter 3, we explore the developing pragmatic and referential communicative abilities of young children using a simplified, child-friendly paradigm. In Chapter 4, we consider the nature of conventions in iterated reference games, in particular to what extent the partner-specificity of these linguistic conventions causes them to be opaque to outsiders. Overall, we find that some signatures of convention formation can occur across a wider variety of circumstances than previously known, and that the descriptions used in iterated reference games are relatively comprehensible to outsiders.

Dedication

To those who supported me in this journey.

Acknowledgments

The path my PhD and my PhD years took was shaped by a number of people to whom I owe thanks for their support.

First, I want to thank my parents for supporting me throughout my entire life. They cultivated intellectualism and knowledge-seeking in me from a young age. They also showed me that going to graduate school (at Stanford) was just a thing one does; the Stanford part is mostly coincidence.

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Chapter 1

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Chapter 2

Introduction: Computational psycholinguistic approaches applied to understanding convention formation in iterated reference games

2.1 Motivations and definitions

The use of language is a feature that distinguishes humans from other species. Language enables coordination between individuals, enforcing social norms via gossip, and communicating complex and abstract ideas, which in turn allows for the cultural accumulation of technologies beyond what any individual could invent. Paired with the technological innovation of writing, language allows for communicating across long distances of space and time.

Humans and other social species can communicate in a variety of modalities, but linguistic structures make language more expressive than other species' communication systems. Humans learn languages in early childhood and then are able to use the shared set of symbol-meaning mappings and combinatorial rules in order to communicate with other humans. Even young children realize that language is a very

powerful system for transmitting ideas from mind to mind, as they demonstrate when they use it to make demands upon their caregivers. As a literate human, language is ubiquitous to (at least my) daily life, but the more one considers language, the more there is to explain and the more impressive our linguistic capabilities seem.

The word “language” can refer to a suite of related concepts all of which are targets of study in related domains. There’s the abstract idea of a specific language, for instance, English as a set of grammatical rules and word meanings. This is an abstract idealization as attempts to specify formal rules that cleanly generate the accepted utterances of a language run up against variability across communities. From a statistical perspective, one can aggregate across variability to study language as it is used by a certain community, not a static entity but something that evolves slowly over time. However, this is also an abstraction, as individuals each have separate grammars that may not fully align even within a community, so meanings must be treated as somewhat probabilistic or distributional. The idea of an individual’s grammar is an abstraction over the process of language learning from individual exposures to speech acts (and texts) and the psychological processes and neural architectures that produce and process language.

Separate from the scale at which we study language, there are other lenses that language is studied through. Language is preserved in artifacts of text and recordings that are studied separately have the human context and intentionality that produced them. Language can also be studied as a capacity for learning a system for expressing ideas and then expressing and understanding ideas in a system. From an information-theoretic lens, language can be viewed as a particular communication channel, subject to the same bounds as any other signal-transmission method. From the opposite perspective, languages often have substantial social and cultural relevance to their users. Generalizing across purposes, language is a tool that humans use to communicate with each other, and we can use both the combinatorial structure of language and pragmatic variations on the structure to accomplish a variety of communicative goals via the meaning conveyed and other features that are implied.

These different aspects of language reveal the complexities of studying language and lead to disagreements around what is studied in linguistics or the “language

sciences”. In the dissertation, I focus primarily on language as a tool for communication, but I will also consider other aspects of language including artifacts of language, language as a shared set of expectations, and psychological processing of language.

Another consideration when studying language use is what counts as language? Some scholarly traditions concentrate on conversational face-to-face language use for communicative purposes, and exclude derivatives like wordplay, phonetic codes, or language models as non-linguistic things that happen to use linguistic symbols. In this work, I do not assign a priori primacy to any particular parts or modalities of language. It may be that some types of language use are easier to process than others, or that some uses of language rely on domain general inferences external to the language faculties, but these are empirical questions whose results I will not assume.

While not as polysemous as the word *language*, *pragmatics* also has a number of partially overlapping meanings. When I use *pragmatics* I mean pragmatics expansively as something along the lines of meaning in context or context as a component to inferring meaning (Bohn et al., 2019).

This dissertation focuses on how humans use language to communicate around the edges of expressivity – how we extend our lexica to accommodate communicative needs, and how we coordinate with each other in these atypical uses of language. These linguistic stretches can be ad hoc and temporary, but if they fill appropriate communicative niches, what are initially atypical usages can spread and lead to the adoption of new words, new meanings and other forms of language change.

As previewed above in discussion of the many angles from which language can be studied, there are trade-offs between studying language in more realistic situations and studying language in more tractable and controlled environments. In this work, I focus on language use *in context* because it is both tractable enough that experiments can usefully inform models and theories and analogous enough to real-world situation to have some explanatory power.

In the rest of the introduction, I first motivate the use of iterated reference games as a model system. I then discuss three main approaches from psycholinguistics that can be applied to convention formation in iterated reference games: efficiency, Bayesian pragmatics, and information theory. Then I review findings from the conversational and iterated reference game literatures and consider how these findings

might be subsumed into larger psycholinguistic frameworks. Finally, I close with avenues for further inquiry and preview the few avenues that are addressed in the rest of the dissertation.

2.2 Iterated reference games as a model system

Iterated reference games are a commonly used paradigm for studying adaptive language use. A *reference game* consist of a set of people communicating with each other about a set of potential referents. One person, the *describer*, has a specific designated target (usually pre-identified as the target by the experimenter) that they seek to communicate to the *matcher* (or matchers). This act of reference requires coordination between describer and matcher on a symbol – meaning alignment. *Iterated* reference games involve repeated reference to the same targets within the same group of people, so that a certain conversational history builds up that could shape the construction and understanding of later symbols. Crucially, this task uses referents that are somehow difficult to name in context, usually by a combination of low-codeability or high similarity to each other.

As described, this task is agnostic to communication modality, and iterated reference games are sometimes conducted using gestures or drawing (Bohn & Frank, 2019; Garrod et al., 2007; Hawkins, Sano, et al., 2023). Here, I focus on how this task interacts with the rich baseline of symbol – meaning mappings afforded by natural language.

People sometimes use words in creative ways to express meanings. For instance, even if a person doesn’t know the name for an object or is having a tip-of-the-tongue event, they can still communicate about the object by using a more circuitous label. If this communicative need occurs repeatedly, an initially ad-hoc referring expression can calcify into a settled name for the object. Communicative acts serve a variety of purposes, but a simple, yet central one is reference: connecting a symbol with a meaning to pick out a target. This is a subgoal needed for more complex acts – it’s hard to tell a story or debate someone without agreement on what different words or phrases refer to.

Combining these observations – that language can be adapted to fill communicative needs, that repeated need leads to changes in referring form, and that reference is a easy to elicit communicative act that is central to other communicative acts – suggests the utility of iterated reference games.

Iterated reference games are a commonly used task. The earliest studies we are aware of are from the 1960s (Krauss & Bricker, 1967; Krauss & Weinheimer, 1964, 1966), and iterated reference games seem to have become more popular starting in the 1980s (Clark & Wilkes-Gibbs, 1986). Iterated reference games have been used to study a wide variety of communicative and linguistic phenomena. The primary targets of study are coordination and the formation of conventions or conceptual pacts [(dahan2025?); Ghaleb et al. (2024); Eliav et al. (2023); Hawkins, Frank, et al. (2020); Metzing & Brennan (2003); Brennan & Clark (1996); among others]. Some studies focus particularly on iterated reference games as a way of studying collaboration in natural conversations (Clark & Wilkes-Gibbs, 1986; Dahan, 2024; Schober & Clark, 1989; Wilkes-Gibbs & Clark, 1992). Other studies focus on other aspects of the coordination, including the negotiation of referring expressions (Bangerter et al., 2020; Knutsen & Le Bigot, 2018) and the role of feedback (Fox Tree & Clark, 2013; Krauss & Glucksberg, 1977; Krauss & Weinheimer, 1966). Iterated reference games with multiple partners are used to study how speakers design their utterances for their specific partners (Horton & Gerrig, 2002, 2005; Turner & Knutsen, 2021; S. O. Yoon & Brown-Schmidt, 2014, 2019a, 2019b) and how speakers might generalize patterns of social conventions across interlocutors (Hawkins, Goodman, et al., 2020; Hawkins et al., 2021). Iterated reference games are used as the dialogue condition in studies on the differences between monologue and dialogue (Branigan et al., 2011; Fox Tree, 1999; Murfitt & McAllister, 2001; **tolins2017?**). Iterated reference games have also been used to study conversation and memory (McKinley et al., 2017; Vanlangendonck et al., 2018), model cultural conflicts between merging teams (Weber & Camerer, 2003), examine speech-gesture tradeoffs (de Ruiter et al., 2012) and study the effects of alcohol on social dynamics (Garrison et al., 2024).

In addition to studies on typical populations, iterated reference games are also used to study the abilities of special populations. Iterated reference games are used with children and teenagers (Branigan et al., 2016; Leung et al., 2024; Turner & Knutsen,

2021) and with older adults (Horton & Spieler, 2007; Hupet et al., 1993; Lysander & Horton, 2012) to study cognitive and linguistic abilities across the lifespan. Iterated reference games are used to study types of memory in people with amnesia (Duff et al., 2005; Duff, Gupta, et al., 2011; Duff, Warren, et al., 2011; S. O. Yoon et al., 2017) and to study various populations with neurological and psychological conditions (Chantraine et al. (1998)).

2.3 Frameworks

Just as one might hope that the results observed in iterated reference games generalize to more realistic situations, one also might hope that the results of iterated reference games could be explained by broad coverage theories of language use. I review three broad approaches that are common within computational psycholinguistics: efficiency, Bayesian pragmatics, and information theory.

2.3.1 Efficiency

One unifying framework in psycholinguistics is efficiency, the idea that language and language use is under pressure to support efficient communication by maximizing the ratio of relevant information transmitted to effort. Efficiency is a high level framework, and as such, requires linking assumptions to enable testing against data. Comparisons between attested languages and counterfactual languages bound what types of linking hypotheses are needed for efficiency to be a parsimonious framework.

Efficiency arises from trade-offs between communicative expressivity and some combination of learnability and ease of production (Kirby et al., 2015; Piantadosi et al., 2012). At any time, a language is thus in a dynamic equilibrium between the usage pressure to communicate many ideas (pushing complexity of language) and some constraints pushing for simplicity, either learnability pressures or ease of production pressures.

Features of language are distributed more closely to the Pareto frontier of the expressivity-simplicity trade-off than would be expected by chance. A well-known observation is that word frequencies follow a power-law distribution, which Zipf (1949) explains in terms of a “principle of least effort”, where words that are more frequent

are also shorter. This type of power-law distribution is common across domains and generated by a variety of processes, so while well known, it is not strong evidence for efficiency as a driver of linguistic distributions (Piantadosi, 2014). Stronger evidence comes from the lexical partitioning of subdomains such as color, number, and kinship terms, where the distribution of systems falls on the Pareto frontier between complexity (number of terms) and informativity (how many bits each term provides) (Gibson et al., 2019; Kemp et al., 2018; Zaslavsky et al., 2018). Syntactic features of language such as harmonic word order and dependency length also appear to be optimized for increased expressivity with minimized processing effort (Gibson et al., 2019; J. Hawkins, 1995).

Arguments for efficiency are based on the language artifacts of grammars and transcripts, but efficiency pressures act on language use as a process, not language as a static code (Gibson et al., 2019). Efficiency imposes a joint constraint on the entire communicative process: minimizing the total time and effort involved in transmitting an idea in one person’s mind to a sufficiently close idea in another person’s mind. While shorter utterances are usually considered more efficient, utterance length in syllables or clock-time is not actually under pressure, but rather the time to produce or parse is under pressure. While length and processing time often correlate, under situations where a verbose alternative is more accessible and comprehensible, it would be favored under efficiency pressures compared to a succinct but hard-to-find wording.

As with other high level frameworks, the parsimony of the efficiency framework is entirely based on its auxiliary linking theories (Gershman, 2018). A way of judging the explanatory adequacy of efficiency is by determining what auxiliary assumptions are needed to connect a given phenomena under the efficiency umbrella and assessing how reasonable are those assumptions are.

As an illustration, I consider what it would take for efficiency to explain ambiguity and redundancy in referring expressions.

Redundancy and over-informative referring expressions

How do we reconcile the idea that language is efficient with violations of efficiency in language use (“look at the yellow banana”) and the ubiquity of ambiguous utterances?

Terms such as “redundant” and “over-informative” are commonly used to describe situations where people produce modifiers that do not restrict the extension of a noun phrase, e.g. “blue cup” when only one cup is salient. Here, the substring “cup” is an alternative utterance that could uniquely identify the target (Rubio-Fernandez et al., 2021). People do produce these non-restricting modifiers in reference tasks, especially for color, but this behavior seems to run counter to the idea of efficient language use.

Formalizing claims of redundancy or over-informativity requires a definition of what would be minimally informative. Defining what is minimally informative requires commitment to a fully specified semantic-pragmatic system, at least for the set of alternative utterances. In the general case, this is difficult to formally specify: for instance, if specificity implicatures are within the option space, are those calculated before or after informativeness is measured (Bergen et al., 2016)? In limited cases, one can sidestep issues of formal theory by empirically measuring the information content of different utterances (Degen et al., 2020), but this does not scale up well in part due to issues with compositionality.

The flip side of “redundancy” is ambiguity: many, many utterances are ambiguous. In general, strong contextual factors render ambiguity a non-issue (Piantadosi et al., 2012), but then the physical and social context must be considered when judging whether language is efficient or redundant. Determining what is efficient language use in-context requires not just analyzing phrases and their alternatives, but also processing and production costs, which may be highly contingent on contextual factors and conversational history.

The idea that utterances should have “just enough” information (Grice, 1989) has inspired a wealth of empirical research into what utterances people produce and what utterances people comprehend. By comparing these two halves of language use, one could determine how calibrated production and comprehension are. Tight alignment between production and comprehension would support an efficiency framework with minimal linking assumptions. In contrast, if discrepancies between what is produced and what is comprehensible, an efficiency framework would require auxiliary assumptions, such as bottlenecks of production, or communication goals other than informativity. One would then have to judge whether these auxiliary assumptions are parsimonious.

2.3.2 Bayesian pragmatics

A broad coverage framework in cognitive science is modeling humans as Bayesian agents making resource rational choices. Linguistic actions can be modeled using Bayesian pragmatic models, where (boundedly) rational Bayesian agents reason about each other to determine the meanings of utterances.

A prominent model within Bayesian pragmatics is the Rational Speech Acts (RSA) framework, an information-theoretic, computational framework for making quantitative predictions about pragmatic inferences in contexts (Frank & Goodman, 2012; Goodman & Frank, 2016). The RSA family of models involves two interlocutors recursively reasoning about how the other would produce or interpret utterances, grounding out in a listener (or speaker) who behaves in a pre-specified “literal” way.

Computational frameworks such as RSA provide a way to factor together different trade-offs and determine their relative weights in a model (Goodman & Frank, 2016). A softmax is taken over the scores of the options to produce a distribution of interpretations and utterances, with some parameters such as the degree of optimality fit based on data. RSA models often use only one or two levels of recursion, which produces a reasonable fit to human experimental judgments, consistent with findings that most people reason pragmatically at a low recursion depth (Franke & Degen, 2016).

RSA models have been used to predict ad hoc pragmatics, conventionalized pragmatic implicatures, and children’s word learning (Bergen et al., 2016; Bohn et al., 2022; Degen et al., 2020; Goodman & Stuhlmüller, 2013; Ohmer et al., 2022). RSA models generally include a utility or informativity term representing how well an utterance resolves uncertainty in favor of the target referent. Other commonly included factors include a saliency prior and some cost function penalizing longer and more complex utterances (Goodman & Frank, 2016). Some models go beyond informativity, incorporating options to infer the question-under-discussion (Kao, 2014; Qing et al., 2016) or balance informativity with politeness (E. J. Yoon et al., 2020).

A full RSA model would incorporate all of these components and infer their weights. However, for tractability, usually only those features that are considered relevant to the domain of interest are included in the model. Because of the flexible framework, RSA can be used to model many sources and levels of uncertainty,

and then integrate out that uncertainty to make predictions, but also update on the sources of uncertainty in response to input.

The Challenge of Semantics

A significant challenge of RSA models is how to define the “literal” listener or speaker. In toy domains, the set of possible utterances is small, and it is feasible to enumerate a set of meanings (Bergen et al., 2016; Frank & Goodman, 2012; Goodman & Stuhlmüller, 2013). A soft or continuous semantics can be used to represent that some dimensions of meaning might be more strongly informative than others, which supports the prediction of patterns of “redundant” color adjective use in referring expressions (Degen et al., 2020).

Soft semantics can conflict with compositionality. Soft semantics can exist without explicit compositionality, provided there is a way to determine match between expression and meaning. At a small scale, this can be determined empirically and fully hand-specified. One option for scaling-up would be neural network system that embeds utterances into semantic space and then taking distances to the target meaning, although this requires making assumptions about what “literal” meaning is. The system might have implicit compositionality, but there are no guarantees about the relationships between the embeddings of “red”, “apple” and “red apple” in these systems. This may be more severe for expressions like “green apple on the red box” and “red apple on the green box” where syntactic relations are key to meaning which sentence-level embeddings may not fully preserve (Erk, 2022). Depending on the domain, embedding systems may or may not have satisfactory match to human semantic judgments.

Alternatively, one could design an explicitly composition system, where meanings are defined on components and there is a process for mapping the meanings of pieces into the meanings of whole phrases. The process could be deterministic rules, or an algorithm or neural model implementation. In formal semantics, a system of logical relations and lambda calculus is used, which can assign that “red apple” is the intersection of “red” and “apple”. However, this formal system would then predict that “apple” is equally true for members of “red apple” and “blue apple”, which doesn’t match people’s judgments due to typicality effects (Degen et al., 2020). The

challenge to both a holistic soft semantics and a compositional approach is that language is language both has strong compositional structure and also cases where the same structure composes meaning differently.

One reason to want both compositionality and soft semantics is to be able to learn from interactions. Given that an expression – meaning pairing was used successfully (or unsuccessfully), one should be able to locally update the lexicon to reflect this additional information, which requires a way to appropriately propagate the updated meaning to component pieces and related phrases.

There is not yet a satisfactory solution for initial or literal meanings in non-toy domains: some situations can be handled by empirically measuring likelihoods in an exhaustive ways, but this holistic approach is not compatible with incremental RSA (Cohn-Gordon et al., 2018) or larger sets of utterances that require compositionality to be defined. Some work has used grounded neural language models for alternative generation and likelihood measures, which has scaling potential in domains that have the right grounded datasets, but not all domains have these datasets (Cohn-Gordon et al., 2018; Monroe & Potts, 2015; White et al., 2020). In order to extend RSA towards more realistic and open-ended scenarios, an important issue is to find a meaning representation that (even at a computational level) will appropriately support pragmatic reasoning.

RSA with history: CHAI

One key obstacle in going from RSA style models to explaining iterated reference games is the need to model the process of transitioning from a successful long, multi-part description to a different, and usually shorter, one. One RSA-style model that can qualitatively generate this pattern of behavior is CHAI, a framework to bridge different levels of convention formation (Hawkins, Franke, et al., 2023).

CHAI uses parameterized hierarchical uncertainty over lexica so that initially, there is variability in how words will be interpreted, which a speaker must integrate over. At the outset, listeners’ lexica are not fully known which represents potential background differences that may be shared by a sub-population and differences from communication history that may be individual-specific. After receiving feedback on the success of the utterance, the new information can be propagated to both the

individual and group lexical representations, resulting in a new prior for future rounds. CHAI can qualitatively account for various convention-formation phenomena in toy systems (Hawkins, Franke, et al., 2023).

CHAI’s explanation for reduction is consistent with the ambiguity arguments of Piantadosi et al. (2012): initially the context is not sufficiently constraining to resolve the ambiguity of a short utterance, so multiple descriptive pieces are needed to triangulate the meaning, but as conversational history accrues, the context is sufficient to disambiguate the shorter description.

In toy models of interlocutors playing a reference game with soft semantics, initial utterances use multiple properties to collectively increase the degree of certainty in the target. This successful reference then shapes the prior over word meanings, until the degree of certainty afforded by only one word is sufficient to pick out a referent.

The same approach CHAI uses to generalize from individuals to groups and populations could also be used to generalize across stimuli. Then, a successful reference (like “man with arms up”) would not only update the meaning of that phrase or components, but could also lend weight to the over-hypothesis about using human and body part based descriptions, and thus update the speaker’s beliefs about the listener’s semantics for the words “head” and “leg”.

As with other RSA models, a primary challenge is how to scale the model and add open-ended semantics to allow for comparison against experimental data using natural language and natural image stimuli.

2.3.3 Information Theory

Information theory is a formalism used in computational psycholinguistics to link abstract ideas to concrete measures and predictions. Information theoretic concepts such as entropy, surprisal, and rate distortion theory have all been key to quantitative theories of language.

Information theory provides a link from functionalist accounts of language usage to the artifacts of language systems, by asking what the constraints are that would render the observed language systems an optimal solution to the functional question of communication (Futrell & Hahn, 2022). One advantage of information theory

is that it provides ways to calculate measures of complexity in a relatively theory-neutral way (Futrell & Hahn, 2022). Measures derived from information theory such as surprisal, conditional entropy, and mutual information have been very productive in language science both for unifying aspects of language typology (ex. dependency locality) (Futrell & Levy, 2017) and for explaining patterns in language processing (ex. surprisal theory and entropy reduction) (Hale, 2001, 2016; Levy, 2008). The idea of rate distortion and a limited channel compressing a meaning through a message has been key to testing efficiency claims related to syntax, word order, and grammatical markers (Futrell & Levy, 2017; Mollica et al., 2021). Information theory based theoretical proofs and toy models have been applied to domains such as language evolution (Plotkin & Nowak, 2000). RSA formulations are closely related to solutions to information-theoretic joint optimization between utility and effort (Zaslavsky et al., 2020).

One difficulty with applying information theoretic measures is the need for a probability distribution over possible meanings and messages; thus, similarly to other frameworks, a potential sticking point is choosing an appropriate distributional language model or datasets to capture the baseline language.

2.4 Phenomena

The communication and conversation literature provides descriptive characterizations of how people communicate in naturalistic and experimental settings. For iterated reference games, the empirical literature provides a fairly consistent and reliable set of results around reduction, convention, and partner-specificity. While there are consistently observable trends in the conversation literature broadly and in iterated reference games specifically, the effects tend to be characterized with verbal theories that posit causes. Research thus far has not rigorously identified necessary and sufficient conditions or dose-response relationships between circumstances (or experimental settings) and the size of the effects. Quantitative accounts for the phenomena are also lacking.

The verbal theories often fail to make explicit predictions about what would happen under conditions other than those tested or what experiments would adjudicate

between theories, which makes testing these theories difficult. Verbal theories often explain their results narrowly as the product of specific situations that are being tested, rather than as part of broader phenomena. As I review the empirical findings and associated theories, I will consider the available evidence, problematize the verbal theories, and consider how broader-coverage approaches from computational psycholinguistics might be used to more formally model or subsume the empirical results.

2.4.1 Mentalizing and non-mentalizing approaches

One historically important distinction in the study of conversation is whether agents track other agents’ internal states. While an intuitively appealing distinction, it would be difficult to measure. One could build, for instance, model-based and model-free algorithms and then compare which better captures an observed phenomenon. This is reminiscent of fights between associationist versus rule-based accounts in linguistics, which don’t seem as relevant now that symbolic and non-symbolic approaches are seen more as different possible implementations. We know now that what looks rule based could be instantiated in a non-symbolic, associationist way, and a rational action symbolic approach could look heuristic under performance load.

Mentalizing tradition and “common ground”

The “mentalizing” tradition treats humans as representing other humans as agents with internal states that include knowledge and goals. Within this umbrella category, approaches vary in how these representations are implemented, how information gets added or modified, what exactly is tracked, and when representations (versus heuristics) are used.

A commonly occurring phrase in the mentalizing tradition is the term “common ground” used to refer to knowledge that two agents share. In some cases, it is used in a pre-theoretic way to mean roughly “things you think another person will understand and won’t be surprised if you reference” (Garrison et al., 2024; Leung et al., 2024). For instance, Hanna et al. (2003) defines common ground as the “mutual knowledge, beliefs, and assumptions” held by the interlocutors. This meaning is roughly comparably to “givenness” in other domains (Fay et al., 2010). However,

“common ground” also has a more specific and theoretically loaded meaning, originating from the privileged versus mutual versus common knowledge framework (Clark & Wilkes-Gibbs, 1986). Under this usage, “common ground” is defined via infinite recursion in knowing that the other person knows that the first person knows that.... This technical usage is used in formal semantics where entities referred to with a definitively article must be introduced to common ground via accommodation, which could itself be seen as a kludge to explain how people get away with making statements whose presuppositions aren’t known to be shared (Horton & Keysar, 1996; Pickering & Garrod, 2004). Empirical work suggests that humans generally don’t engage in more than a couple layers of recursion in their pragmatic reasoning (Franke & Degen, 2016). Thus, it is generally not necessary to distinguish knowledge types at deeper recursion levels than mutual knowledge that both people know to be mutual.

Even if we accept that common ground has grown away from its formal semantic roots to become a synonym of “mutual knowledge”, there is still a question of how agents determine what is mutually known. A common answer is to lay out a deterministic set of rules that supposedly characterize how an agent knows a piece of information is mutually known (Brown-Schmidt, 2012; Clark & Wilkes-Gibbs, 1986; Horton & Keysar, 1996). Some of these rules are generally useful, such as the idea that items in shared visual presence are mutually known, but there are always going to be exceptions and uncertainty, which calls into question the utility of a deterministic flow-chart approach.

The idea of mutual knowledge can be easily subsumed within a Bayesian pragmatics framework. While the “mentalizing” approach places special emphasis on common ground, it is not clear whether there is a reason to model shared knowledge differently from cases where one models another’s knowledge that one does not personally have. For instance, in deciding who to learn from, someone might determine that another agent knows what’s behind a door and then use this information to, for instance, interpret a quantifier like “some” (Goodman & Stuhlmüller, 2013).

Another advantage of the Bayesian approach is that one can avoid deterministic rules and instead reason under uncertainty. Modeling uncertainty could be useful for explaining hedging behavior, and joint updating could be used to model situations

where a piece of knowledge was originally thought to be shared, but new information decreases this likelihood. The framework of flexible inference and probabilistic knowledge states is a powerful tool.

Interactive alignment theory

In contrast to mentalizing approaches, “interactive alignment theory” claims explain how people can successfully collaborate on reference tasks without reasoning about each others’ mental states (Gandolfi et al., 2022; Pickering & Garrod, 2004). Accounts of naive egocentrism suggest that mentalizing is difficult, which provides motivation for an account of collaborative behavior without reference to mental states. Recent work suggests that, contra naive egocentrism, humans reason socially about each other readily and from a young age (Rakoczy, 2022). If mentalizing is not in fact prohibitively costly, the need for a non-mentalizing theory is much reduced. However, some scientists have believed that explicit reasoning about other minds was too computationally expensive and slow to adequately explain behavior, and Pickering & Garrod (2004) tried to account for observed alignment between interlocutors by extending ideas from structural priming.

Structural priming occurs when processing a grammatical structure (for instance, a double object construction) increases the likelihood of expressing an idea using that structure rather than an alternative (producing a double object rather than a prepositional object construction) (Tooley & Traxler, 2010). In production, structural priming is observed even when there is not lexical overlap (ex. different verbs), but is increased when there is lexical overlap. Structural priming in comprehension has been observed only with the same lexical head item (Tooley & Traxler, 2010). The mechanisms of structural priming are an active area of theorizing, without any consensus. The tendency to reuse constructions, styles, or words to express ideas can be observed in non-experimental, naturalistic settings (Doyle et al., 2016; Doyle & Frank, 2016; Pickering & Ferreira, 2008).

Pickering & Garrod (2004) claims that the alignment occurs via “priming” and is “resource-free and automatic”, without providing an explanation of how this would work in terms of memory or processing. Unfortunately, as expressed, interactive alignment theory doesn’t provide an algorithmic or mechanistic level theory. While

inspired from structural priming, interaction alignment goes much further. Structural priming makes narrow claims around why people tend to say “the ball that’s red” after their partner says “the block that’s blue”, whereas interactive alignment claims there is trickle-up and trickle-down effects across linguistic levels that reinforce alignment and align people’s “situation models”. This is a strong claim that is not well supported by a testable algorithm, and does not make risky predictions. The phenomena of interest, repetition, could be explained by some sort of “priming”, or it could be explained by explicit reasoning or production-side ease-of-use constraints (MacDonald, 2013).

Another issue with interactive alignment theory is that it does not account for one of the key features of iterated reference games: convergence to conventionalized referring expressions that evolve and shorten. Interactive alignment does not have an explanation for this pattern of change, rather than mere repetition. Interactive alignment theory generally uses symmetric designs where participants both have partial information, so it may be built as an explanation for a different type of result than the asymmetric describer/matcher designs common to the mentalizing tradition.

While conversational alignment is a worthwhile area of study, the theories of conversational alignment do not provide a satisfying account of the phenomena of iterated reference games. The concern that full inference is not cognitively feasible may be more fruitfully explored using approximate inference in a resource-rational framework or by placing information theoretic limits on the capacity to represent features of individual partners.

2.4.2 Partner specificity

In iterated reference games, describers typically revert to longer descriptions when they interact with a new matcher. This change in behavior is described as “partner-specificity”, as the conventional names developed with one partner are specific to that partnership and not transferred over to other new partners (Brennan & Clark, 1996; Hawkins et al., 2021; Metzing & Brennan, 2003). Partner-specificity is more generally also used to refer to how different pairs diverge to different names for the targets (Hawkins, Frank, et al., 2020). Partner specificity assumes that speakers are mentally representing their partners and their partner’s knowledge states, but is

agnostic to the exact form of representation, and so could be compatible with heuristic or distributed representations in addition to symbolic.

Empirical evidence from experiments where one director talks with multiple partners suggests that people do “partial pooling” over their partners (Hawkins, Goodman, et al., 2020; S. O. Yoon & Brown-Schmidt, 2019a). A speaker A may vary their descriptions when talking to partner B versus partner C, but A talking with B will more like A talking with C than D talking to E, as there is some shared generalization that A makes across B and C. Paired with the tendency to reduce descriptions over time, partial pooling, leads to a jagged pattern of reference length: when switching to a new partner, speakers use longer utterances, but not as long as their initial utterance with their first partner (S. O. Yoon & Brown-Schmidt, 2019a).

A reversion to more elaborate descriptions for new matchers could be efficient if the context from the initial utterances is requisite to understanding the shorter versions. If the elaborated and shortened forms of the concepts (ex “arms up, like a goalie diving for a ball” versus “goalie”) are considered synonyms (in the same way chimp and chimpanzee are synonyms), then the switching could be explained under a uniform information density model, provided that the longer form occurs in a context when it is more surprising and informative (Mahowald et al., 2013).

Partial pooling and partner specificity can be instantiated in a hierarchical model of Bayesian pragmatics, such as CHAI or lexical uncertainty models (Bergen et al., 2016; Hawkins, Franke, et al., 2023; Potts et al., 2015; Schuster & Degen, 2020). These models provide theoretic coverage, but testing these frameworks against open-lexicon natural language data is not yet computationally tractable.

From an information-theoretic lens, the mutual information would include the conversation history, the current utterance, and the visual context of targets. Within a partnership, the conversation history can serve as an encoder that allows later conversational elements to use a more efficient coding to map between language and referent. Partner specificity effects could arise when, conditioned on conversation history, the target and convention predict each other, but target and convention do not predict each other given no history or a different history.

2.4.3 Audience design & effort sharing

Audience design is the idea that speakers appear sensitive to the knowledge state of their listener and say things that are easy for the listeners to comprehend. Broadly understood, audience design is occurring when a person speaks differently to a 5-year old than to an adult, or to a colleague than to a lay person. However, despite the intuition that audience design occurs in everyday interactions, it is difficult to determine if intentional audience design occurs in iterated reference games.

The phrase “audience design” sometimes implies intention on the part of the speaker (Horton & Gerrig, 2005; **horton2002a?**), and sometimes is used when utterances are constructed based on what’s easy for the speaker, and listener ease is a side effect (Horton & Keysar, 1996; MacDonald, 2013; Rogers et al., 2013). For instance, giving an elaborated description to a naive listener could be an intentional design choice (that contrasts with a shorter describer for a non-naive listener), but a speaker could also start descriptions with given material because it is more accessible to them, and only incidentally convenient for the listener. Audience design is sometimes used even more broadly to refer to production choices that make utterances easier to comprehend (V. S. Ferreira, 2019). In a very broad sense, efficiency pressures for uniform information density could be construed as audience design.

Speakers and listeners often share recent context, find the same things salient, and linguistically what is easier to produce is often easier to process. Thus, disentangling ease of production from ease of comprehension requires careful experimental settings where these two processes diverge (F. Ferreira, 2004).

Audience design relates to the broader topic of how interlocutors split the communicative burden with one another. Depending on the task and the communication modality, speakers and listeners have different options for how to balance the communicative load (Clark & Wilkes-Gibbs, 1986; Fay et al., 2010; Fox Tree & Clark, 2013). For instance, a listener could describe what options they see or otherwise prompt the speaker. Load splitting may vary based on the capacities of the interlocutors (ex. a speaker might craft their utterances more when talking to a child versus an adult) and the capacities of the channels (ex. speakers may use different approaches if listeners can interrupt).

Multi-way conversations complicate the theories of audience design and partner

specificity by introducing a larger audience of more partners. Two main questions are whether speakers “aim low” or “aim high” in balancing the needs of the listeners and whether speakers track listeners as individuals or an aggregate (S. O. Yoon & Brown-Schmidt, 2017, 2019a).

Empirical results indicate that speakers are sensitive to the individual knowledge states of listeners. Describers seem to track the knowledge states of different matchers independently when they alternate between addressing matchers (S. O. Yoon & Brown-Schmidt, 2017). When simultaneously addressing a group of matchers with disparate knowledge levels, speakers give descriptions that are fairly similar to what they give naive listeners (S. O. Yoon & Brown-Schmidt, 2017). However, there is also some gradient, where descriptions vary based on the proportion of naive listeners in groups of up to 4 addressees (S. O. Yoon & Brown-Schmidt, 2019a). With pairs of listeners who differ both in their background knowledge and their visual context, speakers produce descriptions that are sensitive to the interaction between background knowledge and distinctiveness of the target in context (S. O. Yoon & Brown-Schmidt, 2019b). Speakers in group contexts sometimes produce utterances to support multiple listeners, such as referring to a target with both the established name and an elaborated description (S. O. Yoon & Brown-Schmidt, 2019a). Research on balancing multiple listeners has not extended to larger groups, so it is unknown how the tracking of partner’s knowledge states might degrade in larger groups.

Larger group communication is instead studied more in network structures, where (non-linguistic) convention formation has been studied on networks of up to 50 people (Guilbeault et al., 2021). In referring to targets sampled from a continuous space, large groups tended to develop fairly consistent category boundaries across independent networks, while small networks had more idiosyncratic category boundaries. This work suggests there may be group-size and group/network-structure dependencies on what sort of conventions arise, but network games on traditional iterated reference games have focused on small networks of four people (Hawkins, Goodman, et al., 2020; Hawkins et al., 2021).

Partner specificity and audience design accounts implicitly assume that the speaker’s behavior is close to optimal. Generally, humans are fairly pragmatic in their communication, so we might expect speakers to be well-calibrated on what listeners will

understand. Speaker behavior certainly reflects the notion that different descriptions are appropriate depending on the listener’s knowledge state of their shared conversation history.

However, direct tests of whether descriptions are optimally tailored require some sort of counterfactual. One option is to test how well naive matchers do given descriptions from different circumstances. Naïve matchers do better the more their observation history resembles that of the original conversation – when hearing descriptions in order instead of in reverse order (Murfitt & McAllister, 2001), when listening to the entire game instead of starting in the third round (Schober & Clark, 1989), and when seeing yoked trials from a single game rather than trials sampled across 10 games (Hawkins, Sano, et al., 2023). Descriptions produced in dialogue are easier to understand than those produced in monologue (Murfitt & McAllister, 2001). The gradient of performance suggests that some changes in descriptions may be adaptive to listeners who share more background. However, naive matchers still have far above chance performance indicating that descriptions are not exclusively understandable by the knowledgeable listeners they were intended for. To the extent that new people can understand later utterances without context, a counterfactual more rapid adoption of short forms could achieve a certain level of accuracy just as well as the observed pattern. Thus, from a purely audience design listener-focused perspective, speakers are clearly not efficient. However, we do not know what the optimal utterance sequences would be for listeners.

Before we are forced to conclude that humans are miscalibrated in their linguistic utterances, we can consider some potential auxiliary hypotheses that could render observed behavior efficient. One possibility is that the communicative bottleneck is firmly on the production side: speakers are doing the best they can to rapidly produce descriptions, and producing more succinct descriptions earlier would be too effortful for speakers. Another possibility is that social norms and task dynamics dissuade the use of short form descriptions even if they are likely to be understood: perhaps the initial descriptions are more polite, or perhaps people are (overly) cautious in their efficiency/accuracy trade-offs. Both of these hypotheses could be tested empirically.

Bayesian pragmatic models could be extended to cover communicating with multiple listeners at once. Rather than just communicate pragmatically with one person,

balancing multiple people’s needs would require balancing different goals. A lexical uncertainty Bayesian pragmatics model could be used to capture the naive listener behavior in eavesdropper scenarios. Effort splitting could in theory be represented by changing RSA parameters such as depth of reasoning, starting lexical semantics, or cost functions.

2.4.4 Convention formation

A key observation from iterated reference games is that dyads tend to form shared “conventions” (also called “conversational pacts”) about how to refer to the initially ambiguous targets. These conventions tend to be partner- and context-specific: changes in the speaker, audience members, or context can all license the use of a new description (Ibarra & Tanenhaus, 2016; Metzing & Brennan, 2003; S. O. Yoon & Brown-Schmidt, 2014). Absent these changes, pairs tend to stick near the semantic space of successful descriptions.

Convention formation and conceptual pacts could occur on different granularities of semantic alignment. Partners could align on the lexical level, by both calling a figure “ballerina”. They could align on conceptualizing the figure as a ballet dancer with a tutu, manifesting in descriptions with semantic, but not lexical overlap, such as “ballerina” and “dancing in a tutu”. Partners could agree on a general paradigm for how to describe figures, such as in terms as humans in different postures. (horton2002a?) distinguish between “lexical entrainment” when the same words are reused, and “conceptual similarity” when there is broader similarity that does not repeat the same words. These levels often co-occur, but in order to computationally quantify the phenomenon of convention formation, we have to make choices about how to operationalize the similarity: is it word overlap, lemma overlap, distance in semantic space, or something else?

In addition to convention formation, repeated references also tend to change from descriptions of a set of features to more holistic labels. Typically, in convention formation, holistic or analogic descriptions are the ones that stick (Clark & Wilkes-Gibbs, 1986), but almost any type of label can sometimes crystallize into the convention, including ones that pick up on low-level or meta-level features. The range of successful options makes explaining convention formation difficult, and suggests that there is

strong path-dependency in the evolution of reduced utterances, potentially influenced by hard to quantify factors such as relationships and humor value.

The length and meaning of an utterance are not a priori related, but these two features tend to correlate in iterated reference games (Hawkins, Frank, et al., 2020). The reduction of utterances is sometimes used as a measurement proxy for the semantic changes (Clark & Wilkes-Gibbs, 1986; Garrison et al., 2024; Hawkins, Franke, et al., 2023), and Bovet et al. (2023) considers it a measure of common ground. Convention and reduction are also conflated with partner-specificity, presumably again because of co-occurrence (Clark & Wilkes-Gibbs, 1986; Wilkes-Gibbs & Clark, 1992). Better tools for measuring each of these constructs more directly and independently will be needed to address the theoretical issues around how interrelated these concepts are. It remains an empirical question whether the shortening of utterances, convention formation, and partner-specificity of descriptions are inseparable or merely occur together in the experimental paradigms considered in the literature.

From an information-theoretic perspective, convention formation is convergence towards an encoding that is more efficient at representing the specific set of target meanings than the baseline language. These meanings are not commonly needed and so start with low codeability, but over local usage, a (temporary) warp of the meanings can be achieved that is locally more optimized to express the target meanings.

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2.4.5 Developmental work

One angle to understanding linguistic phenomena is to look at their developmental trajectory to understand when the phenomena arises and how it changes across development.

There have been a limited number of direct studies of iterated references games with children, although there is a much wider body of work on young children’s communicative and pragmatic abilities.

One influential early study suggested that 4-5-year-old preschoolers struggle with child-child referential communication (Glucksberg et al., 1966). In their paradigm, one child was given a set of 6 blocks in a specific order. Their task was to describe the image on each block so their partner could pick out their corresponding block.

As children described and selected blocks, they stacked them on pegs. While 4-5-year-old children succeeded on practice trials with familiar shapes and visual access to each other's blocks, children failed on critical trials where the blocks had abstract drawings and there was no visual access. Even after multiple rounds with the same images, children were not able to correctly order the blocks. Glucksberg et al. (1966) attributed children's communicative failures to their production of ego-centric descriptions that did not account for the other child's perspective. Similar experiments with older children indicated a gradual improvement through adolescence both for initial accuracy and for the increase in accuracy across repetitions. Still, even the 9th grade sample was noticeably worse than the adult college student sample (Glucksberg & Krauss, 1967; Krauss & Glucksberg, 1969).

Since Glucksberg et al. (1966), few studies have revisited the question of whether children can successfully communicate in an iterated reference game. In one study, 8-10-year-olds exhibited adult-like patterns of increasing accuracy, increasing speed, and shorter descriptions across repetitions, but children's accuracy was still far below adult performance and highly variable between dyads (Branigan et al., 2016). 4-6-year-olds successfully used conventionalized gestures with their partners in an iterated reference game where children could only use gestures to communicate (Bohn & Frank, 2019). 4-8-year-old children succeeded at playing an iterated reference game with a parent using a simple, child-friendly tablet-based task (Leung et al., 2024). In this task, even 4-year-olds had an initial accuracy above 80%, which rose to above 90% in later repetitions (Leung et al., 2024).

One issue that arises in developmental work is assessing whether the paradigms are measuring the abilities of interest or whether task demands or cognitive limitations are masking nascent abilities. The overall trend in developmental work has been discovering evidence of abilities emerging earlier than previously thought, due to better methods and designs that require fewer additional cognitive resources. Thus, we might be skeptical of whether the negative results of Glucksberg et al. (1966) are an accurate reflection of children's communicative abilities. The perspective taking abilities of 4 to 5 year olds do seem to be greater than previously thought (San Juan et al., 2025).

What could psycholinguistic frameworks add to our understanding of developmental pragmatics? If we had a computational model of adult performance, the question would be what aspects to change or restrict to approximate child behavior. For an RSA type model, we could consider a lower optimization parameter, or reducing the amount of computation (sampling fewer options). Children might also do a similar process, but with a different size and baseline semantics in the lexicon. Would their updates in a repeated model be smaller or bigger? Even if we can't fit the model, the modeling formalism suggests ways of thinking about the differences between children and adults in a dimensional way.

If we take efficiency to mean an optimal tradeoff between expressivity and effort, then children might have different weightings between the two, ending up on a different frontier or a different part of the frontier. For instance, cognitive capacity constraints, or things like working memory or executive control could mean that certain types of utterances are particularly effortful, and thus shift the optimal curve to include less expressive utterances. For child – adult dyads, the shift in speaker / listener capacity may be substantially different than for adult - adult dyads, rendering different approaches more efficient. This may be difficult to test since it would require figuring how where the bottlenecks are empirically, or having good enough models to compare fits to the data.

From an information theory lens, we could conceive of communication channels involving children as being extra noisy or having a more limited channel capacity or both.

2.4.6 Incremental approaches to processing and production

Another angle from which to view reference is at the level of fine grained time course. As a referring expression is incrementally processed, what is the listener doing? How early do effects related to speaker identity or pragmatics occur?

The typical approach to understanding the unfolding of comprehension in real time is via visual world eye tracking where looks are interpreted as meaning to potential targets. Alternative methods to get at incremental interpretation and processing difficulty can also be used.

One idea that visual world approaches to reference try to address is whether there

is an initial stage of interpretation that differs from the final resolution. For instance, certain influences like the literal meaning of the utterance might be available sooner than pragmatic or more contextualized interpretations. This is sometimes characterized as “top-down” versus “bottom-up” but there may also be different sources such as linguistic context or social context or visual access.

The relative balance and time course of top-down and bottom-up influences has been a big question in psycholinguistics and neurolinguistics for a long time (Gwilliams & Davis, 2022; Horton & Gerrig, 2005; Horton & Keysar, 1996; Tanenhaus et al., 1995). One debate in particular, in the (non-iterated) reference game literature specifically concerned the role of egocentrism in the early stages of linguistic processing. One side of the debate was the Egocentric Anchoring and Adjustment theory, where Keysar et al. (2000) argued for an initially egocentric perspective on the basis that people often initially look at objects that are good matches to a description even if the object is not mutually visible to the speaker Keysar et al. (2003). In particular, this theory claims a two-stage model where there is an initial (completely) egocentric perspective that can later be overcome, in contrast to the (much less controversial) idea that there is a general egocentric bias where one of the sources of information is egocentric but it can be in parallel to theory of mind information (Rubio-Fernández, 2008). The ego-centric first claim rests on a couple dubious linking hypotheses: that if people consider an interlocuter’s perspective, their prior should be that the interlocuter only refers to mutually known things; and separately, that looking at an object is a sign that it is considered a potential referent (eye-tracking data is widely interpreted this way, but there is evidence that this proxy is only approximate, Degen et al., 2021). Their studies also always had the egocentric-only lure be extremely salient, by making it by far the best match to the description.

The counterpoint to initial ego-centrism presented by Hanna et al. (2003) is a constraint-based theory where many factors can play into comprehension. Many factors may influence language production and comprehension, to varying extents, and on differing time courses. Their studies use egocentric lures that are equivalently good to the shared visibility target and find early perspective taking, and confirm this with other experiments in different paradigms (Hanna et al., 2003; Nadig & Sedivy, 2002).

The larger picture of language processing and neuroscience strongly suggests that context information (at least from linguistic context) is used to predict upcoming linguistic material, and thus that “top-down” information can affect even early stages of linguistic processing such as phoneme identification and lexical processing (Gwilliams et al., 2023; Gwilliams & Davis, 2022). Predictive coding is a theory with good coverage of neural evidence in many domains, including language, with language processing regions seemingly sensitive to both lexical and structural information in predicting upcoming words (Shain et al., 2020). However, we might still wonder if earlier behavioral readouts are more influenced by perceptual information and less pragmatic, or if cognitive load also sways the balance of influence.

2.4.7 Production constraints

Compared to the theoretic landscape around psycholinguistic processing, production is much less studied because it’s a lot harder to study experimentally.

There is agreement that production requires conceptualization, formulation, and articulation, but that is about all that is agreed upon (Bürki, 2018). From interference studies, we can get some sense of when certain elements in sentences are planned (Momma et al., 2016). On a word level, the phonological realization of words that are more frequent or usually appear in more predictable contexts is reduced relative to words that usually convey more information (Bürki, 2018). This could be consistent with either a production facilitation account or an audience design account – speaker and listener language statistics are similar so it is difficult to disentangle. For production beyond the scale of single words, that are just more questions, including what the planning looks like, how the time courses of different stages of production may overlap, and what size and types of units are stored in memory (Bürki, 2018). There’s a lot we don’t know about the mechanics of production even in situations with less context-sensitivity than iterated reference games.

As with behavioral measures, neuroscience measures are also used more for processing than for production, and articulatory processes are better studied than the higher level planning processes. fMRI studies indicate that language production tends to activate the same regions as language processing (unsurprisingly), but these regions are especially active during phrase structure building, suggesting that production may

have higher costs than comprehension (Hu et al., 2022). However, fMRI does not have precise time course resolution, so it can’t resolve questions of how information flows in the brain.

The increased load from production is consistent with accounts that suggest that utterance design (and language evolution) are substantially shaped by biases to ease production. MacDonald (2013) points out three production biases: easy first, plan reuse, and reduce interference that could explain some patterns of utterance production, including in reference games.

One approach to identifying speaker-side difficulties in iterated reference games is to look for disfluencies (S. O. Yoon & Brown-Schmidt, 2014). Intuitively, producing descriptions of abstract figures is difficult and there may be load based on task difficulty that leads to non-optimal productions (ex using “square head” as part of a description when it is not diagnostic) that are reminiscent of children’s describe/identify difficulties.

There are approaches one could take to try to understand more what production-side pressures are, including time-course locked measures such as eye-tracking and disfluency analysis. Carefully controlled studies and comparisons to monologue conditions may also help. Varying time pressure or pre-articulation planning time could also be used to help sort out what’s going on (Horton & Keysar, 1996). But as the larger questions around production generally show, this is hard to study and may require very careful and clever experiment designs.

What do the broader coverage frameworks have to say about production? Production-side constraints are one of the factors posited to drive linguistic efficiency, but there aren’t fleshed out predictions about speakers would or wouldn’t say. In general RSA operates on a level of abstraction that only considers entire utterances and so only considers production cost in terms of the utterance sampling procedure (if not all options are considered) and the costs assigned to each utterance. However, given that most iterated reference games allow the speaker to keep talking, one could (again, in theory) adapt incremental RSA (Cohn-Gordon et al., 2018) to explain potential order effects within what parts of descriptions are produced first.

One information-theoretic computational-level model of incremental language production comes from Futrell (2023). This model applies the Rate Distortion Theory of

Control to production, suggesting a trade-off between incremental steps that are good for the communicative goal (ex. being informative) with what is easy to produce, that is what is more automatic and predictable. As with other information-theoretic models, it has good explanations in toy or qualitative domains for explaining production ordering preferences, the use of filled pauses, and good-enough word production.

2.5 Where does this leave us?

A core theoretical question is whether the observed patterns of reduction, convention, and partner-specificity are due to specific mechanisms that only operate in a limited scope of iterated-reference-game-like situations or whether these results are explainable by broader coverage theories of pragmatic language use.

The phenomena observed in iterated reference games do not appear under other circumstances where people use pragmatic reasoning (including regular reference games). One possibility is that there are mechanisms specific to certain circumstances, and these mechanisms lead to the observed phenomena. For instance, some theories may focus only on conversational coordination with common ground and a backchannel. The exact scopes are often ambiguous, but they are not trying to account for all language use or even all pragmatic referential language productions. In some cases, the implicit scope may be smaller than the range of circumstances where the phenomena is observed, as is the case for theories that focus on dyadic mechanisms.

Alternatively, it could be that the same mechanisms that account for pragmatics broadly can also account for the phenomena in iterated reference games. The particular phenomena of interest in iterated reference games arise from general mechanisms playing out in certain circumstances such as when targets have low codeability and reference is repeated to the same target with the same partner. These circumstances are incorporated into the model as a parameters (what the targets are, what the contextualized lexicon is, what the communication channel is) and certain combinations of the parameters lead to the emergence of things like partner-specific convention formation.

Most of the iterated reference game literature has been focused on descriptive

results and developing small-scope theories. Here I’ve been examining what sorts of parameterizations and auxiliary hypotheses would be needed for broad-scope theories to account for the results of iterated reference games. I believe our default hypothesis should be that convention formation emerges from the same mechanisms that account for all the rest of language use. As we attempt to figure out what set of mechanisms account for language use, iterated reference games are an interesting model for highly contextual language production and comprehension.

Having surveyed the phenomena and small-scope theories of iterated reference games, I now return to the three frameworks and synthesize how they might apply and what I think the interesting open questions are.

2.5.1 Efficiency

Sometimes, judging whether something is efficient or not may be clear cut, if all reasonable sets of assumptions return the same result. It is perhaps easier to judge something to be inefficient if there is a shorter alternative that is both easier to produce and easier to comprehend. In the general case, however, efficiency is very hard to cache out in specific predictions. What’s efficient for an utterance in isolation may not be efficient when considered over an entire life of language use.

The formation of conversational pacts is often characterized as efficient, but this claim has not been cached out in formal models (Clark & Wilkes-Gibbs, 1986; Hawkins, Frank, et al., 2020). “Efficiency” is an informal description of these pacts that captures that shorter expressions are *more* efficient, but a given shorted expression could still be mis-calibrated to a particular situation. It could be too short, leading to misunderstanding, or still longer than it needs to be for effective communication. Mis-calibration of this type could contradict claims about partner specificity of reduction, but would not have to. Communicators could intend to be optimally calibrated but run into production difficulties, masking their competence. Either way, current claims of efficiency in conversation do not connect directly with the formal claims of optimal efficiency in the “language design” literature, and it is unknown whether speakers and listeners are calibrated or not.

Even with a specified task goal (communicate the target to a partner), what we tell people and what their actual utility function is are not the same. It’s still

a conversation between people so communication norms may also play a role, and that's difficult to control while retaining naturalism.

There are some open questions that are testable, mostly focused on figuring out what sorts of linking hypotheses would be required for the behavior to be (mostly) efficient.

One is figuring out how understandable utterances are from different points in the conversation, to basically test the idea that speakers are reducing “as fast as possible” for listeners; that is, both that extra words in an utterance are adding utility (adding information, adding certainty), and that the context of earlier utterances (and their repetition) is necessary for understanding later utterances (i.e. reducing faster would negatively impact accuracy). These are addressable questions, both with models and information theory applied to utterances and with testing new matchers on manipulated samples or orders of utterances.

The other issue is trying to get a handle on production-side demands. If speakers have the hard task, then perhaps what they say is mostly driven by ease of planning. This is harder to test, but one could do monologue studies and dialog studies with eye-tracking and time course measures to try to get at it in more fine grained detail.

Generally efficiency is usually applied to language the culturally evolved system based out of pressures that apply at each step. But it doesn't actually require that the individual steps be fully at the frontier, merely that there be enough pressure towards expressivity and ease overall that the system shifts in this direction. So, even if it turns out that people really don't seem efficient at this task, it doesn't hurt the efficiency framework that much because this task may be substantially out of distribution and thus, people's behavior on it is not shaping language use on a large scale.

2.5.2 Bayesian pragmatics

Bayesian pragmatics is a very promising large-scale framework for explaining a lot of linguistic behavior in context and incorporating history and non-linguistic context into a model of utterance production and comprehension. The main challenge is figuring out how to scale up to handle the natural language that comes up in reference games.

There are also some modeling considerations for how to extend models to deal with different types of data – how to deal with multiple listeners simultaneously, how to abstract across items to learn conceptual paradigms (i.e. abstracting from items to an approach of describing images in terms of what the person is doing). One question that comes up with Bayesian pragmatic models is how many free parameters they have and whether one is actually getting a parsimonious model with broad coverage, or just a set of related models very tuned to individual datasets. This is a broad issue in modeling and the closer we can get to a generative or cognitive model (even if fairly abstracted) the better.

So one way of making progress and seeing how broadly Bayesian models can be applied would be extending the existing models for a wider variety of designs. Group communication, either in the form of one speaker - multiple listeners (S. O. Yoon & Brown-Schmidt, 2014, 2019a, 2019b) or network structures (Hawkins, Goodman, et al., 2020), poses a challenge as there is not yet a non-dyadic extension of RSA that can do partial pooling across listeners and jointly optimize for the responses of multiple interlocutors. It should be possible to design extensions to CHAI to accommodate different group structures.

However, it would still lead to questions of whether the small-scale models can predict the patterns seen in natural language, full scale experiments. And that requires solving problems related to the open vocabulary. How do you set baseline (literal) meanings for any utterance someone might produce, and how do you allow those to update in a sensible way? With toy models, one can do composition semantics and update based on the compositionality. While we could ground semantics in some neural model (ex. in some sort of VLM) that doesn't give composition units that can be individually updated. Thus, I think some of the challenge of getting a robust test of Bayesian pragmatics models requires a lot of NLP and ML engineering.

CHAI, while not a full explanation for the full patterns of real-world data, is a big step forward in at least providing a framework that actually explains why reduction would be optimal. It remains to be seen whether RSA-style models can predict the slope of reduction and the content of words that stay versus drop, but I believe the attempt would push forward our understanding of pragmatics and communication. Even without a cached out model, this approaches can inform how we think about

experiments and results.

2.5.3 Information theory

Unlike efficiency and Bayesian pragmatics which are frameworks, information theory is more of a tool. It's robust to see that in iterated reference games, the utterances tend to get shorter, but what's actually going on with the language? In order to quantify that, it's useful to have some sort of metrics to use. One could theorize that words that are more informative are more likely to stick around and become the conventions. To test this, one would need a measure of "information" to apply, which may now be possible to test using computational models of language and stimuli.

Information theory also provides measures for trying to understand how context supports understanding. This could be to formalize ideas of common ground by considering the joint information between the conversation history (as the common ground) and the current utterance. Again, to measure this at scale we would need models to approximate the information, but there are now models that could be used.

The idea of task relevant information also makes this an interesting place to apply linguistic measures of information theory. For normal psycholinguistic texts, information theoretic predictors like surprisal or entropy reduction have substantial predictive value on processing difficulty as indexed by measures of reading time (or neural signals) (Smith & Levy, 2013; Wilcox et al., 2021). While surprisal is a measure of information, we might think that it's not the same as how many useful bits of information are conveyed. (Gibberish is quite high surprisal, but low information.)

One way that information theory might usefully frame what happens in iterated reference games is that the language is adapting to the specific task of optimally coding the target options. An optimal code would only need 3.5 bits to express each of the 12 options (a common number of options in iterated reference games). Many more bits than this of natural language are used in early rounds – suggesting that initially standard natural language is not a good fit for this task. Some of this language is conveying task-relevant information, just in a wordy way, while other parts may be not-task relevant at all (ex: describing features that do not vary across targets). Over time, people probably both drop some of the filler and identify the task-relevant components, but they also develop a more efficient code for those features. This

formulation makes some testable predictions about the entropy of distributions over utterances at different points during the game.

2.5.4 Summary

From this survey of approaches as well as phenomena and theories, I think there are a few takeaways. I’ve highlighted some places where further work is needed to answer the question of how well broad-coverage psycholinguistic approaches explain iterated reference game results. Some of this is completely outside the reference domain – we could do with a better understanding of (general) language planning and production, and we could benefit from natural language processing models that do compositionality that could be plugged into Bayesian frameworks to interface with real data. There are also some computational and modeling directions, highlighted above.

One other direction is better characterizing the **explanandum**: what exactly are the phenomena we are hoping to characterize? While there have been many studies of iterated reference games across some variety of conditions, some theoretically relevant aspects of the space are under-explored. Some of the reference game specific theories focus on narrow parts of iterated reference game space, while all the broad coverage approaches expect a mostly continuous relationship between circumstantial knobs and the outcome measures. From a modeling perspective, having more data and a wider range of data also reduces some concerns that we are over-fitting to a very narrow circumstance.

Thus, I believe one necessity to substantial progress in the iterated reference game domain is a better empirical and descriptive understanding of the phenomena of interest. How do circumstantial knobs (group size, communication modality, stimuli, etc.) push around the extend to which we observe reduction, convention formation, and partner specificity? We don’t have to finely map the entire landscape, but some sense of the geography of experimental space is necessary (Almaatouq et al., 2022).

More variable experimental circumstances could also address the question of how closely related the different phenomena of interest are. At least where it’s been studied, reduction, semantic alignment within a group, and partner specificity have patterned together (to the point where some metrics are used as proxies for each

other). How closely are they actually linked? Does reduction only occur via convention formation? A greater range of experimental situations could tell how well these measures separate versus correlate in different situations, which will in turn clarify what theory should explain.

2.6 Outline of the dissertation

In this dissertation, we expand the empirical foundation of iterated reference games in three directions, with the aim of addressing some theoretical assumptions and questions and providing some data for computational modeling in the future. We also apply some computational models to survey coarse-grained patterns of changes in semantic similarities.

In Chapter 2, we address the generalization patterns of the key patterns of results in iterated reference games, by examining the trajectories of convention formation in games of varying small group sizes and varying communication channel capacities. This expands the empirical foundation to mimic more of the different circumstances in which communication occurs in the real world, and allows for a test of how entangled the different reduction phenomena really are.

In Chapter 3, we turn to the developmental timeline of the communication and coordination skills embedded in iterated reference tasks. Here we re-visit historical claims that children are too egocentric to produce successful ad-hoc referential expressions, which is fairly incompatible with more recent work showing nascent pragmatic language use and communication skills in preschoolers. This different context again provides a test of how entangled different observed reduction phenomena really are and provides insights into how language use fits into broader cognitive and linguistic development.

Finally, in Chapter 4, we measure the level of opacity in the referential expressions produced in iterated reference games (using the corpus of reference games from Chapter 2). Taking inspiration from lexical uncertainty models, we use both naive human matchers and computational proxies to determine how far from “baseline semantics” referential expressions are at different stages of convention formation. These patterns are a test of at least the strongest versions of efficiency and audience design claims

and give us a better understanding of what sort of partner-specificity is at play.

In the last chapter, I conclude with a discussion of where this leaves us and what directions seem most promising from here.

References

Placeholder

- Almaatouq, A., Griffiths, T. L., Suchow, J. W., Whiting, M. E., Evans, J., & Watts, D. J. (2022). Beyond playing 20 questions with nature: Integrative experiment design in the social and behavioral sciences. *Behavioral and Brain Sciences*, 47, e33. <https://doi.org/10.1017/S0140525X22002874>
- Bangerter, A., Mayor, E., & Knutsen, D. (2020). Lexical entrainment without conceptual pacts? Revisiting the matching task. *Journal of Memory and Language*, 114, 104129. <https://doi.org/10.1016/j.jml.2020.104129>
- Bergen, L., Levy, R., & Goodman, N. (2016). Pragmatic reasoning through semantic inference. *Semantics and Pragmatics*, 9(20), 1–91. <https://doi.org/10.3765/sp.9.20>
- Bohn, M., & Frank, M. C. (2019). The Pervasive Role of Pragmatics in Early Language. *Annual Review of Developmental Psychology*, 1(1), 223–249. <https://doi.org/10.1146/annurev-devpsych-121318-085037>
- Bohn, M., Kachel, G., & Tomasello, M. (2019). Young children spontaneously recreate core properties of language in a new modality. *Proceedings of the National Academy of Sciences*, 116(51), 26072–26077. <https://doi.org/10.1073/pnas.1904871116>
- Bohn, M., Schmidt, L. S., Schulze, C., Frank, M. C., & Tessler, M. H. (2022). Modeling Individual Differences in Children’s Information Integration During Pragmatic Word Learning. *Open Mind*, 6, 311–326. https://doi.org/10.1162/opmi_a_00069
- Bovet, V., Knutsen, D., & Fossard, M. (2023). Direct and indirect linguistic measures of common ground in dialogue studies involving a matching task: A systematic

- review. *Psychonomic Bulletin & Review*, 31(1), 122–136. <https://doi.org/10.3758/s13423-023-02359-2>
- Branigan, H. P., Bell, J., & McLean, J. F. (2016). Do You Know What I Know? The Impact of Participant Role in Children’s Referential Communication. *Frontiers in Psychology*, 7. <https://doi.org/10.3389/fpsyg.2016.00213>
- Branigan, H. P., Catchpole, C. M., & Pickering, M. J. (2011). What makes dialogues easy to understand? *Language and Cognitive Processes*, 26(10), 1667–1686. <https://doi.org/10.1080/01690965.2010.524765>
- Brennan, S. E., & Clark, H. H. (1996). Conceptual pacts and lexical choice in conversation. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 22(6), 1482–1493. <https://doi.org/10.1037/0278-7393.22.6.1482>
- Brown-Schmidt, S. (2012). Beyond common and privileged: Gradient representations of common ground in real-time language use. *Language and Cognitive Processes*, 27(1), 62–89. <https://doi.org/10.1080/01690965.2010.543363>
- Bürki, A. (2018). Variation in the speech signal as a window into the cognitive architecture of language production. *Psychonomic Bulletin & Review*, 25(6), 1973–2004. <https://doi.org/10.3758/s13423-017-1423-4>
- Champagne-Lavau, M., Fossard, M., Martel, G., Chapdelaine, C., Blouin, G., Rodriguez, J.-P., & Stip, E. (2009). Do patients with schizophrenia attribute mental states in a referential communication task? *Cognitive Neuropsychiatry*, 14(3), 217–239. <https://doi.org/10.1080/13546800903004114>
- Chantraine, Y., Joannette, Y., & Ska, B. (1998). Conversational abilities in patients with right hemisphere damage. *Journal of Neurolinguistics*, 11(1-2), 21–32. [https://doi.org/10.1016/S0911-6044\(98\)00003-7](https://doi.org/10.1016/S0911-6044(98)00003-7)
- Clark, H. H., & Wilkes-Gibbs, D. (1986). Referring as a collaborative process. *Cognition*, 22(1), 1–39. [https://doi.org/10.1016/0010-0277\(86\)90010-7](https://doi.org/10.1016/0010-0277(86)90010-7)
- Cohn-Gordon, R., Goodman, N. D., & Potts, C. (2018). An Incremental Iterated Response Model of Pragmatics. *arXiv:1810.00367 [Cs]*. <https://doi.org/10.48550/arxiv.1810.00367>
- Dahan, D. (2024). When hedging helps, rather than impedes, communication: Collaboration in the referential communication task. *Discourse Processes*, 62(2), 89–111. <https://doi.org/10.1080/0163853X.2024.2437738>

- de Ruiter, J. P., Bangerter, A., & Dings, P. (2012). The Interplay Between Gesture and Speech in the Production of Referring Expressions: Investigating the Tradeoff Hypothesis. *Topics in Cognitive Science*, 4(2), 232–248. <https://doi.org/10.1111/j.1756-8765.2012.01183.x>
- Degen, J., Hawkins, R. D., Graf, C., Kreiss, E., & Goodman, N. D. (2020). When redundancy is useful: A Bayesian approach to “overinformative” referring expressions. *Psychological Review*, 127(4), 591–621. <https://doi.org/10.1037/rev0000186>
- Degen, J., Kursat, L., & Leigh, D. (2021). Seeing is believing: Testing an explicit linking assumption for visual world eye-tracking in psycholinguistics. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 43.
- Doyle, G., & Frank, M. C. (2016). Investigating the Sources of Linguistic Alignment in Conversation. *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 526–536. <https://doi.org/10.18653/v1/P16-1050>
- Doyle, G., Yurovsky, D., & Frank, M. C. (2016). A Robust Framework for Estimating Linguistic Alignment in Twitter Conversations. *Proceedings of the 25th International Conference on World Wide Web*, 637–648. <https://doi.org/10.1145/2872427.2883091>
- Duff, M. C., Gupta, R., Hengst, J. A., Tranel, D., & Cohen, N. J. (2011). The Use of Definite References Signals Declarative Memory. *Psychological Science*, 22(5), 666–673. <https://doi.org/10.1177/0956797611404897>
- Duff, M. C., Hengst, J., Tranel, D., & Cohen, N. J. (2005). Development of shared information in communication despite hippocampal amnesia. *Nature Neuroscience*, 9(1), 140–146. <https://doi.org/10.1038/nn1601>
- Duff, M. C., Warren, D. E., Gupta, R., Vidal, J. P. B., Tranel, D., & Cohen, N. J. (2011). Teasing apart tangrams: Testing hippocampal pattern separation with a collaborative referencing paradigm. *Hippocampus*, 22(5), 1087–1091. <https://doi.org/10.1002/hipo.20967>
- Eliav, R., Ji, A., Artzi, Y., & Hawkins, R. D. (2023). *Semantic uncertainty guides the extension of conventions to new referents* (No. arXiv:2305.06539). arXiv. <https://doi.org/10.48550/arXiv.2305.06539>

- Erk, K. (2022). The Probabilistic Turn in Semantics and Pragmatics. *Annual Review of Linguistics*, 8(1), 101–121. <https://doi.org/10.1146/annurev-linguistics-031120-015515>
- Fay, N., Garrod, S., Roberts, L., & Swoboda, N. (2010). The Interactive Evolution of Human Communication Systems. *Cognitive Science*, 34(3), 351–386. <https://doi.org/10.1111/j.1551-6709.2009.01090.x>
- Ferreira, F. (2004). Production-comprehension asymmetries. *Behavioral and Brain Sciences*, 27(02), 196–196. <https://doi.org/10.1017/S0140525X04280050>
- Ferreira, V. S. (2019). A Mechanistic Framework for Explaining Audience Design in Language Production. *Annual Review of Psychology*, 70(1), 29–51. <https://doi.org/10.1146/annurev-psych-122216-011653>
- Fox Tree, J. E. (1999). Listening in on monologues and dialogues. *Discourse Processes*, 27(1), 35–53. <https://doi.org/10.1080/01638539909545049>
- Fox Tree, J. E., & Clark, N. B. (2013). Communicative Effectiveness of Written Versus Spoken Feedback. *Discourse Processes*, 50(5), 339–359. <https://doi.org/10.1080/0163853X.2013.797241>
- Frank, M. C., & Goodman, N. D. (2012). Predicting Pragmatic Reasoning in Language Games. *Science*, 336(6084), 998–998. <https://doi.org/10.1126/science.1218633>
- Franke, M., & Degen, J. (2016). Reasoning in Reference Games: Individual- vs. Population-Level Probabilistic Modeling. *PLOS ONE*, 11(5), e0154854. <https://doi.org/10.1371/journal.pone.0154854>
- Futrell, R. (2023). Information-theoretic principles in incremental language production. *Proceedings of the National Academy of Sciences*, 120(39), e2220593120. <https://doi.org/10.1073/pnas.2220593120>
- Futrell, R., & Hahn, M. (2022). Information Theory as a Bridge Between Language Function and Language Form. *Frontiers in Communication*, 7. <https://doi.org/10.3389/fcomm.2022.657725>
- Futrell, R., & Levy, R. (2017). Noisy-context surprisal as a human sentence processing cost model. In M. Lapata, P. Blunsom, & A. Koller (Eds.), *Proceedings of the 15th Conference of the European Chapter of the Association for Computational*

- Linguistics: Volume 1, Long Papers* (pp. 688–698). Association for Computational Linguistics.
- Gandolfi, G., Pickering, M. J., & Garrod, S. (2022). Mechanisms of alignment: Shared control, social cognition and metacognition. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 378(1870), 20210362. <https://doi.org/10.1098/rstb.2021.0362>
- Garrison, A. C. S., Yoon, S. O., Brown-Schmidt, S., Ariss, T., & Fairbairn, C. E. (2024). Alcohol and common ground: The effects of intoxication on linguistic markers of shared understanding during social exchange. *Psychology of Addictive Behaviors*, 38(1), 79–91. <https://doi.org/10.1037/adb0000922>
- Garrod, S., Fay, N., Lee, J., Oberlander, J., & MacLeod, T. (2007). Foundations of Representation: Where Might Graphical Symbol Systems Come From? *Cognitive Science*, 31(6), 961–987. <https://doi.org/10.1080/03640210701703659>
- Gershman, S. J. (2018). How to never be wrong. *Psychonomic Bulletin & Review*, 26(1), 13–28. <https://doi.org/10.3758/s13423-018-1488-8>
- Ghaleb, E., Rasenberg, M., Pouw, W., Holler, J., & Fernandez, R. (2024). *Analysing Cross-Speaker Convergence in Face-to-Face Dialogue through the Lens of Automatically Detected Shared Linguistic Constructions*. <https://doi.org/10.48550/arXiv.2405.08546>
- Gibson, E., Futrell, R., Piantadosi, S. P., Dautriche, I., Mahowald, K., Bergen, L., & Levy, R. (2019). How Efficiency Shapes Human Language. *Trends in Cognitive Sciences*, 23(5), 389–407. <https://doi.org/10.1016/j.tics.2019.02.003>
- Glucksberg, S., & Krauss, R. M. (1967). What do people say after they have learned how to talk? Studies of the development of referential communication. *Merrill-Palmer Quarterly of Behavior and Development*, 13(4), 309–316. <https://www.jstor.org/stable/23082551>
- Glucksberg, S., Krauss, R., & Weisburg, R. (1966). *Referential Communication in Nursery School Children: Method and Some Preliminary Findings*.
- Goodman, N. D., & Frank, M. C. (2016). Pragmatic Language Interpretation as Probabilistic Inference. *Trends in Cognitive Sciences*, 20(11), 818–829. <https://doi.org/10.1016/j.tics.2016.08.005>
- Goodman, N. D., & Stuhlmüller, A. (2013). Knowledge and Implicature: Modeling

- Language Understanding as Social Cognition. *Topics in Cognitive Science*, 5(1), 173–184. <https://doi.org/10.1111/tops.12007>
- Grice, H. P. (1989). Logic and conversation. In *Studies in the way of Words*. Academic Pr.
- Guilbeault, D., Baronchelli, A., & Centola, D. (2021). Experimental evidence for scale-induced category convergence across populations. *Nature Communications*, 12(1), 327. <https://doi.org/10.1038/s41467-020-20037-y>
- Gupta, R., Tranel, D., & Duff, M. C. (2012). Ventromedial prefrontal cortex damage does not impair the development and use of common ground in social interaction: Implications for cognitive theory of mind. *Neuropsychologia*, 50(1), 145–152. <https://doi.org/10.1016/j.neuropsychologia.2011.11.012>
- Gwilliams, L., & Davis, M. H. (2022). Extracting Language Content from Speech Sounds: The Information Theoretic Approach. In L. L. Holt, J. E. Peelle, A. B. Coffin, A. N. Popper, & R. R. Fay (Eds.), *Speech Perception* (pp. 113–139). Springer International Publishing. https://doi.org/10.1007/978-3-030-81542-4_5
- Gwilliams, L., Marantz, A., Poeppel, D., & King, J.-R. (2023). Top-down information shapes lexical processing when listening to continuous speech. *Language, Cognition and Neuroscience*, 39(8), 1045–1058. <https://doi.org/10.1080/23273798.2023.2171072>
- Hale, J. (2001). A probabilistic earley parser as a psycholinguistic model. *Second Meeting of the North American Chapter of the Association for Computational Linguistics on Language Technologies 2001 - NAACL '01*, 1–8. <https://doi.org/10.3115/1073336.1073357>
- Hale, J. (2016). Information-theoretical Complexity Metrics. *Language and Linguistics Compass*, 10(9), 397–412. <https://doi.org/10.1111/lnc3.12196>
- Hanna, J. E., Tanenhaus, M. K., & Trueswell, J. C. (2003). The effects of common ground and perspective on domains of referential interpretation. *Journal of Memory and Language*, 49(1), 43–61. [https://doi.org/10.1016/S0749-596X\(03\)00022-6](https://doi.org/10.1016/S0749-596X(03)00022-6)
- Hawkins, J. (1995). *A Performance Theory of Order and Constituency*. Cambridge University Press.

- Hawkins, R. D., Frank, M. C., & Goodman, N. D. (2020). Characterizing the Dynamics of Learning in Repeated Reference Games. *Cognitive Science*, 44(6). <https://doi.org/10.1111/cogs.12845>
- Hawkins, R. D., Franke, M., Frank, M. C., Goldberg, A. E., Smith, K., Griffiths, T. L., & Goodman, N. D. (2023). From partners to populations: A hierarchical Bayesian account of coordination and convention. *Psychological Review*, 130(4), 977–1016. <https://doi.org/10.1037/rev0000348>
- Hawkins, R. D., Goodman, N. D., Goldberg, A. E., & Griffiths, T. L. (2020). Generalizing meanings from partners to populations: Hierarchical inference supports convention formation on networks. *arXiv:2002.01510 [Cs]*. <https://doi.org/10.48550/arxiv.2002.01510>
- Hawkins, R. D., Liu, I., Goldberg, A. E., & Griffiths, T. G. (2021). Respect the code: Speakers expect novel conventions to generalize within but not across social group boundaries. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 43.
- Hawkins, R. D., Sano, M., Goodman, N. D., & Fan, J. E. (2023). Visual resemblance and interaction history jointly constrain pictorial meaning. *Nature Communications*, 14(1). <https://doi.org/10.1038/s41467-023-37737-w>
- Horton, W. S., & Gerrig, R. J. (2002). Speakers’ experiences and audience design: Knowing when and knowing how to adjust utterances to addressees. *Journal of Memory and Language*, 47(4), 589–606. [https://doi.org/10.1016/S0749-596X\(02\)00019-0](https://doi.org/10.1016/S0749-596X(02)00019-0)
- Horton, W. S., & Gerrig, R. J. (2005). The impact of memory demands on audience design during language production. *Cognition*, 96(2), 127–142. <https://doi.org/10.1016/j.cognition.2004.07.001>
- Horton, W. S., & Keysar, B. (1996). When do speakers take into account common ground? *Cognition*, 59(1), 91–117. [https://doi.org/10.1016/0010-0277\(96\)81418-1](https://doi.org/10.1016/0010-0277(96)81418-1)
- Horton, W. S., & Spieler, D. H. (2007). Age-related differences in communication and audience design. *Psychology and Aging*, 22(2), 281–290. <https://doi.org/10.1037/0882-7974.22.2.281>
- Hu, J., Small, H., Kean, H., Takahashi, A., Zekelman, L., Kleinman, D., Ryan, E.,

- Nieto-Castañón, A., Ferreira, V., & Fedorenko, E. (2022). Precision fMRI reveals that the language-selective network supports both phrase-structure building and lexical access during language production. *Cerebral Cortex*, 33(8), 4384–4404. <https://doi.org/10.1093/cercor/bhac350>
- Hupet, M., Chantraine, Y., & Nef, F. (1993). References in conversation between young and old normal adults. *Psychology and Aging*, 8(3), 339–346. <https://doi.org/10.1037/0882-7974.8.3.339>
- Ibarra, A., & Tanenhaus, M. K. (2016). The Flexibility of Conceptual Pacts: Referring Expressions Dynamically Shift to Accommodate New Conceptualizations. *Frontiers in Psychology*, 7. <https://doi.org/10.3389/fpsyg.2016.00561>
- Kao, J. T. (2014). Formalizing the Pragmatics of Metaphor Understanding. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 36.
- Kemp, C., Xu, Y., & Regier, T. (2018). Semantic Typology and Efficient Communication. *Annual Review of Linguistics*, 4(1), 109–128. <https://doi.org/10.1146/annurev-linguistics-011817-045406>
- Keysar, B., Barr, D. J., Balin, J. A., & Brauner, J. S. (2000). Taking Perspective in Conversation: The Role of Mutual Knowledge in Comprehension. *Psychological Science*, 11(1), 32–38. <https://doi.org/10.1111/1467-9280.00211>
- Keysar, B., Lin, S., & Barr, D. J. (2003). Limits on theory of mind use in adults. *Cognition*, 89(1), 25–41. [https://doi.org/10.1016/S0010-0277\(03\)00064-7](https://doi.org/10.1016/S0010-0277(03)00064-7)
- Kirby, S., Tamariz, M., Cornish, H., & Smith, K. (2015). Compression and communication in the cultural evolution of linguistic structure. *Cognition*, 141, 87–102. <https://doi.org/10.1016/j.cognition.2015.03.016>
- Knutsen, D., & Le Bigot, L. (2018). The influence of conceptual (mis)match on collaborative referring in dialogue. *Psychological Research*, 84(2), 514–527. <https://doi.org/10.1007/s00426-018-1060-1>
- Krauss, R. M., & Bricker, P. D. (1967). Effects of Transmission Delay and Access Delay on the Efficiency of Verbal Communication. *The Journal of the Acoustical Society of America*, 41(2), 286–292. <https://doi.org/10.1121/1.1910338>
- Krauss, R. M., & Glucksberg, S. (1969). The Development of Communication: Competence as a Function of Age. *Child Development*, 40(1), 255. <https://doi.org/10.2307/1127172>

- Krauss, R. M., & Glucksberg, S. (1977). Social and Nonsocial Speech. *Scientific American*, 236(2), 100–105. <https://doi.org/10.1038/scientificamerican0277-100>
- Krauss, R. M., & Weinheimer, S. (1964). Changes in reference phrases as a function of frequency of usage in social interaction: A preliminary study. *Psychonomic Science*, 1(1-12), 113–114. <https://doi.org/10.3758/BF03342817>
- Krauss, R. M., & Weinheimer, S. (1966). Concurrent feedback, confirmation, and the encoding of referents in verbal communication. *Journal of Personality and Social Psychology*, 4(3), 343–346. <https://doi.org/10.1037/h0023705>
- Leung, A., Yurovsky, D., & Hawkins, R. D. (2024). Parents spontaneously scaffold the formation of conversational pacts with their children. *Child Development*, 96(2), 546–561. <https://doi.org/10.1111/cdev.14186>
- Levy, R. (2008). A Noisy-Channel Model of Human Sentence Comprehension under Uncertain Input. *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing*, 234–243.
- Lysander, K., & Horton, W. S. (2012). Conversational Grounding in Younger and Older Adults: The Effect of Partner Visibility and Referent Abstractness in Task-Oriented Dialogue. *Discourse Processes*, 49(1), 29–60. <https://doi.org/10.1080/0163853X.2011.625547>
- MacDonald, M. C. (2013). How language production shapes language form and comprehension. *Frontiers in Psychology*, 4. <https://doi.org/10.3389/fpsyg.2013.00226>
- Mahowald, K., Fedorenko, E., Piantadosi, S. T., & Gibson, E. (2013). Info/information theory: Speakers choose shorter words in predictive contexts. *Cognition*, 126(2), 313–318. <https://doi.org/10.1016/j.cognition.2012.09.010>
- McKinley, G. L., Brown-Schmidt, S., & Benjamin, A. S. (2017). Memory for conversation and the development of common ground. *Memory & Cognition*, 45(8), 1281–1294. <https://doi.org/10.3758/s13421-017-0730-3>
- Metzing, C., & Brennan, S. (2003). When conceptual pacts are broken: Partner-specific effects on the comprehension of referring expressions. *Journal of Memory and Language*, 49(2), 201–213. [https://doi.org/10.1016/S0749-596X\(03\)00028-7](https://doi.org/10.1016/S0749-596X(03)00028-7)

- Mollica, F., Bacon, G., Zaslavsky, N., Xu, Y., Regier, T., & Kemp, C. (2021). The forms and meanings of grammatical markers support efficient communication. *Proceedings of the National Academy of Sciences*, 118(49), 135–138. <https://doi.org/10.1073/pnas.2025993118>
- Momma, S., Slevc, L. R., & Phillips, C. (2016). The timing of verb selection in Japanese sentence production. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 42(5), 813–824. <https://doi.org/10.1037/xlm0000195>
- Monroe, W., & Potts, C. (2015). Learning in the Rational Speech Acts Model. *arXiv:1510.06807 [Cs]*. <https://doi.org/10.48550/arxiv.1510.06807>
- Murfitt, T., & McAllister, J. (2001). The Effect of Production Variables in Monolog and Dialog on Comprehension by Novel Listeners. *Language and Speech*, 44(3), 325–350. <https://doi.org/10.1177/00238309010440030201>
- Nadig, A. S., & Sedivy, J. C. (2002). Evidence of Perspective-Taking Constraints in Children’s On-Line Reference Resolution. *Psychological Science*, 13(4), 329–336. <https://doi.org/10.1111/j.0956-7976.2002.00460.x>
- Ohmer, X., Franke, M., & König, P. (2022). Mutual Exclusivity in Pragmatic Agents. *Cognitive Science*, 46(1), e13069. <https://doi.org/10.1111/cogs.13069>
- Piantadosi, S. T. (2014). Zipf’s word frequency law in natural language: A critical review and future directions. *Psychonomic Bulletin & Review*, 21(5), 1112–1130. <https://doi.org/10.3758/s13423-014-0585-6>
- Piantadosi, S. T., Tily, H., & Gibson, E. (2012). The communicative function of ambiguity in language. *Cognition*, 122(3), 280–291. <https://doi.org/10.1016/j.cognition.2011.10.004>
- Pickering, M. J., & Ferreira, V. S. (2008). Structural priming: A critical review. *Psychological Bulletin*, 134(3), 427–459. <https://doi.org/10.1037/0033-2909.134.3.427>
- Pickering, M. J., & Garrod, S. (2004). Toward a mechanistic psychology of dialogue. *Behavioral and Brain Sciences*, 27(02). <https://doi.org/10.1017/S0140525X04000056>
- Plotkin, J. B., & Nowak, M. A. (2000). Language Evolution and Information Theory. *Journal of Theoretical Biology*, 205(1), 147–159. <https://doi.org/10.1006/jtbi.2000.2053>

- Potts, C., Lassiter, D., Levy, R., & Frank, M. C. (2015). Embedded Implicatures as Pragmatic Inferences under Compositional Lexical Uncertainty. *Journal of Semantics*, ffv012. <https://doi.org/10.1093/jos/ffv012>
- Qing, C., Goodman, N. D., & Lassiter, D. (2016). A rational speech-act model of projective content. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 38.
- Rakoczy, H. (2022). Foundations of theory of mind and its development in early childhood. *Nature Reviews Psychology*, 1(4), 223–235. <https://doi.org/10.1038/s44159-022-00037-z>
- Rogers, S. L., Fay, N., & Maybery, M. (2013). Audience Design through Social Interaction during Group Discussion. *PLoS ONE*, 8(2), e57211. <https://doi.org/10.1371/journal.pone.0057211>
- Rubio-Fernandez, P., Mollica, F., & Jara-Ettinger, J. (2021). Speakers and listeners exploit word order for communicative efficiency: A cross-linguistic investigation. *Journal of Experimental Psychology: General*, 150(3), 583–594. <https://doi.org/10.1037/xge0000963>
- Rubio-Fernández, P. (2008). On the automaticity of egocentricity: A review of the Egocentric Anchoring and Adjustment model of perspective taking. *UCL Working Papers in Linguistics*, 20.
- San Juan, V., Gibbard, K., Morra, S., Khu, M., Chambers, C. G., & Graham, S. A. (2025). Common ground without copresence: Preschoolers apply indirect cues about shared knowledge in real-time referential interpretation. *Cognitive Development*, 74, 101573. <https://doi.org/10.1016/j.cogdev.2025.101573>
- Schober, M. F., & Clark, H. H. (1989). Understanding by addressees and overhearers. *Cognitive Psychology*, 21(2), 211–232. [https://doi.org/10.1016/0010-0285\(89\)90008-X](https://doi.org/10.1016/0010-0285(89)90008-X)
- Schuster, S., & Degen, J. (2020). I know what you’re probably going to say: Listener adaptation to variable use of uncertainty expressions. *Cognition*, 203, 104285. <https://doi.org/10.1016/j.cognition.2020.104285>
- Shain, C., Blank, I. A., van Schijndel, M., Schuler, W., & Fedorenko, E. (2020). fMRI reveals language-specific predictive coding during naturalistic sentence comprehension. *Neuropsychologia*, 138, 107307. <https://doi.org/10.1016/j.neuropsychologia.>

- 2019.107307
- Smith, N. J., & Levy, R. (2013). The effect of word predictability on reading time is logarithmic. *Cognition*, 128(3), 302–319. <https://doi.org/10.1016/j.cognition.2013.02.013>
- Tanenhaus, M. K., Spivey-Knowlton, M. J., Eberhard, K. M., & Sedivy, J. C. (1995). Integration of Visual and Linguistic Information in Spoken Language Comprehension. *Science*, 268(5217), 1632–1634. <https://doi.org/10.1126/science.7777863>
- Tooley, K. M., & Traxler, M. J. (2010). Syntactic Priming Effects in Comprehension: A Critical Review. *Language and Linguistics Compass*, 4(10), 925–937. <https://doi.org/10.1111/j.1749-818X.2010.00249.x>
- Turner, C., & Knutsen, D. (2021). Audience Design in Collaborative Dialogue between Teachers and Students. *Discourse Processes*, 58(8), 703–725. <https://doi.org/10.1080/0163853X.2021.1904768>
- Vanlangendonck, F., Takashima, A., Willems, R. M., & Hagoort, P. (2018). Distinguishable memory retrieval networks for collaboratively and non-collaboratively learned information. *Neuropsychologia*, 111, 123–132. <https://doi.org/10.1016/j.neuropsychologia.2017.12.008>
- Weber, R. A., & Camerer, C. F. (2003). Cultural Conflict and Merger Failure: An Experimental Approach. *Management Science*, 49(4), 400–415. <https://doi.org/10.1287/mnsc.49.4.400.14430>
- White, J., Mu, J., & Goodman, N. D. (2020). Learning to refer informatively by amortizing pragmatic reasoning. *arXiv:2006.00418 [Cs]*. <https://doi.org/10.48550/arxiv.2006.00418>
- Wilcox, E., Vani, P., & Levy, R. (2021). A Targeted Assessment of Incremental Processing in Neural Language Models and Humans. *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, 939–952. <https://doi.org/10.18653/v1/2021.acl-long.76>
- Wilkes-Gibbs, D., & Clark, H. H. (1992). Coordinating beliefs in conversation. *Journal of Memory and Language*, 31(2), 183–194. [https://doi.org/10.1016/0749-596X\(92\)90010-U](https://doi.org/10.1016/0749-596X(92)90010-U)

- Yoon, E. J., Tessler, M. H., Goodman, N. D., & Frank, M. C. (2020). Polite Speech Emerges From Competing Social Goals. *Open Mind*, 4, 71–87. https://doi.org/10.1162/opmi_a_00035
- Yoon, S. O., & Brown-Schmidt, S. (2014). Adjusting conceptual pacts in three-party conversation. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 40(4), 919–937. <https://doi.org/10.1037/a0036161>
- Yoon, S. O., & Brown-Schmidt, S. (2017). Aim Low: Mechanisms of Audience Design in Multiparty Conversation. *Discourse Processes*, 55(7), 566–592. <https://doi.org/10.1080/0163853X.2017.1286225>
- Yoon, S. O., & Brown-Schmidt, S. (2019a). Audience Design in Multiparty Conversation. *Cognitive Science*, 43(8), e12774. <https://doi.org/10.1111/cogs.12774>
- Yoon, S. O., & Brown-Schmidt, S. (2019b). Contextual Integration in Multiparty Audience Design. *Cognitive Science*, 43(12), e12807. <https://doi.org/10.1111/cogs.12807>
- Yoon, S. O., Duff, M. C., & Brown-Schmidt, S. (2017). Learning and using knowledge about what other people do and don’t know despite amnesia. *Cortex*, 94, 164–175. <https://doi.org/10.1016/j.cortex.2017.06.020>
- Zaslavsky, N., Hu, J., & Levy, R. P. (2020). *A Rate-Distortion view of human pragmatic reasoning* (No. arXiv:2005.06641). arXiv. <https://doi.org/10.48550/arXiv.2005.06641>
- Zaslavsky, N., Kemp, C., Regier, T., & Tishby, N. (2018). Efficient compression in color naming and its evolution. *Proceedings of the National Academy of Sciences*, 115(31), 7937–7942. <https://doi.org/10.1073/pnas.1800521115>
- Zipf, G. (1949). *Human behavior and the principle of least effort: An introduction to human ecology*.